

CHAPTER 8 - TESTING AND RESULTS

8.1 Introduction

Testing ensures that every module of EduGenie functions correctly and meets the project requirements.

8.2 Testing Objectives

Verify functionality, validate inputs, test backend APIs, confirm frontend-backend communication, and evaluate AI-generated outputs.

8.3 Functional Testing

Question Answering, Explanation, Quiz Generation, Summary, and Learning Path Recommendation were tested successfully with sample inputs.

8.4 User Interface Testing

The interface was tested for page loading, navigation, responsiveness, input validation, and output display.

8.5 Backend Testing

FastAPI endpoints were verified to ensure correct request handling and AI integration.

8.6 Results

The application successfully generated answers, explanations, quizzes, summaries, and learning recommendations through the web interface.

8.7 Performance Analysis

The system provided fast responses, stable execution, and an intuitive user experience.

8.8 Chapter Summary

Testing confirmed that EduGenie satisfies its functional requirements and operates reliably.